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EDUCATION

Shandong University, Qingdao

2024 – 2028

Institute of Frontier and Interdisciplinary Science · class of 2024, expected 2028

Coursework: theoretical mechanics, electrodynamics, thermodynamics and statistical physics, quantum mechanics, advanced quantum mechanics, quantum field theory, particle physics, general relativity.

HOW I WORK

Derive from first principles, not from precedent — ugly is fine, fabricated is not. Every claim has to replay from a log.

RESEARCH INTERESTS

I am interested in how much of hadron structure can be derived rather than fitted.

Light-front QCD. Quantise on the light front and the vacuum turns trivial — the difficulty does not disappear, it reappears in the Fock truncation and in the zero modes the $A^+=0$ gauge leaves behind. Next for me: the holographic link from confinement scale to bound-state equation.

Hadron structure from Fock wave functions. A transverse-momentum distribution read as a spectator-integrated amplitude, not a fitting function, with $\Sigma x_i=1$ and $\Sigma \mathbf{k}_{\perp i}=0$ built in from the start.

TMD factorisation. Collins–Soper evolution and its non-perturbative kernel; the SIDIS structure-function dictionary (Sivers, Collins, Boer–Mulders); the Drell–Yan sign reversal that keeps SIDIS and DY from being fitted apart.

RESEARCH DIRECTIONS & WORK IN PROGRESS

- **Light-front quantisation (PH 01).** Working toward: an answer to where the Fock-space truncation can still be trusted. The two handles I am following are DLCQ matrices whose UV cutoff is matched to the truncation, and what the zero modes keep carrying once the gauge is fixed.
- **Baryon light-cone Fock expansion (PH 02).** Working toward: a truncation checkable against constraints that do not bend — normalisation ($\Sigma_n \int |\psi_n|^2=1$, $\int dx\, f_1$ equal to the valence count) and the mass relation $P^+P^- - P_\perp^{\,2}=M^2$ — three-quark sector first, then five, and which observables survive a cut at fixed N.
- **Proton structure — DIS / SIDIS / DY (PH 03).** Working toward: one picture the three processes agree on. TMD factorisation at $P_{hT} \ll Q$, matched onto a Y-term at large transverse momentum, Collins–Soper evolution with a non-perturbative kernel, and where a Gaussian width in $\langle k_\perp{}^2 \rangle$ gives way to power-law tails.

CURRENT WORK

- **PH 02 · Baryon light-front wave function at fixed N.** three-quark sector first, then how far the five-quark component moves the structure function.
- **PH 03 · Cross-process constraints.** DIS, SIDIS and DY in one parameterisation, starting where they disagree.

OPEN QUESTIONS

- **PH 01 · Light-front quantization.** Does the multiplicative kinematics actually save work, or does it just move the work into the Fock space? · If the vacuum got simpler, which physics got moved into the wave function instead? · Which observable is the right monitor for truncation error — and how would I know, before I have the exact answer?
- **PH 02 · Baryon light-cone Fock expansion.** Is convergence driven by cutting $k\perp$ or by adding Fock components? · Which observables are least sensitive to the cut — and are they the interesting ones? · In which process do five-quark components rise above the truncation error?
- **PH 03 · Proton structure — DIS / SIDIS / DY.** Do SIDIS and DY disagree on the same $\langle k\perp^2 \rangle$ — and is a disagreement a sign reversal or an evolution effect? · In which Q^2 range does the Gaussian width model start to fail? · Which term of the 3D distribution is shortest on data — and is that a fact about the data, or about the factorisation?

TECHNICAL SKILLS

LANGUAGES	Python for numerics and fits, C++ for compute, Bash, Git
SYMBOLIC	Mathematica for algebra; SymPy — mathkit, an open-source skill I maintain; SymPy for Agent (upcoming); numerical integration and fitting; ROOT for plots and data; convolution and evolution codes
THEORY	Light-front quantisation, Fock-basis expansion, TMD factorisation and evolution, PDF and TMD fits, perturbative calculations
TOOLING	LaTeX and TikZ, reproducible notes, Linux and HPC, papers and data management

LANGUAGES

Chinese (native) · English (working proficiency).